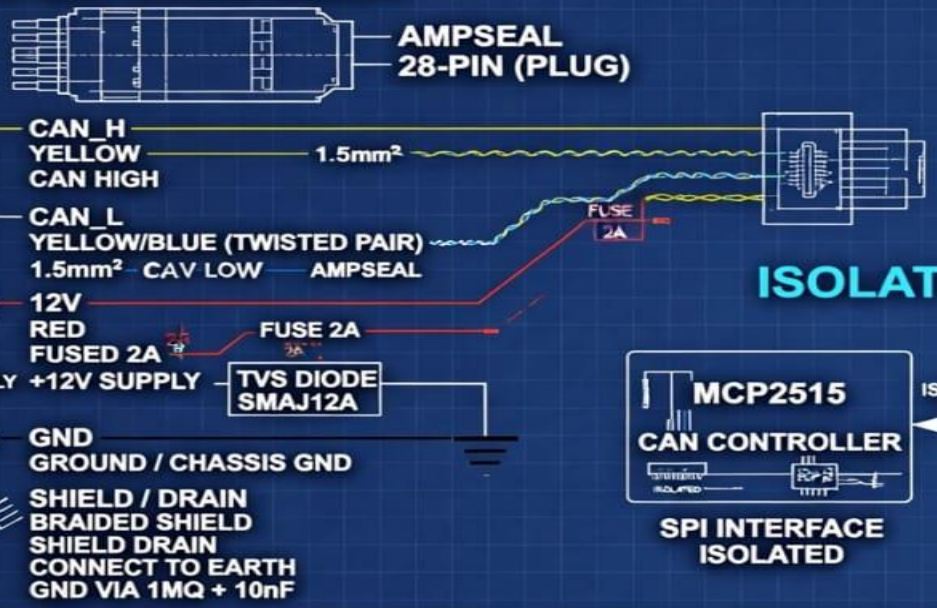
# KIA EV3 PESQUERIA - WIRING HARNESS DIAGRAM

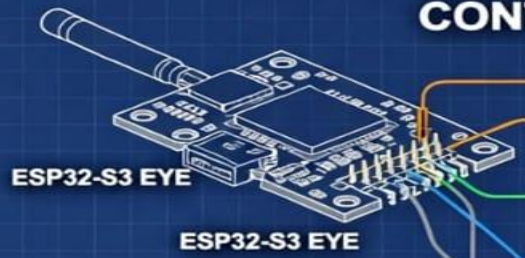
ENGINEER: Norberto Bravo Tadeo, Cheran Michoacan

DRAWING NO: KIA-EV3-PESQ-WH-001 REV B

## 28-PIN AMPSEAL CONNECTOR — KIA SPEC 18790-02850




## VCU ESP32-S3 EYE CONTROL MODULE




SCK — PIN 18 ORANGE 0.5mm² TXL to ESP32-S3 GPIO18

MISO — GREEN 0.5mm² TXL ESP32-S3 GPIO19

MOSI — PIN 23 to ESP32-S3 GPIO23

CS — GRAY to ESP32-S3 GPIO5

## ISOLATED CAN BUS INTERFACE




SPI INTERFACE ISOLATED

2.5kV ISOLATION BARRIER

OPTO-ISOLATION INCLUDED

## SIGNAL TABLE — WIRING & SPECIFICATIONS

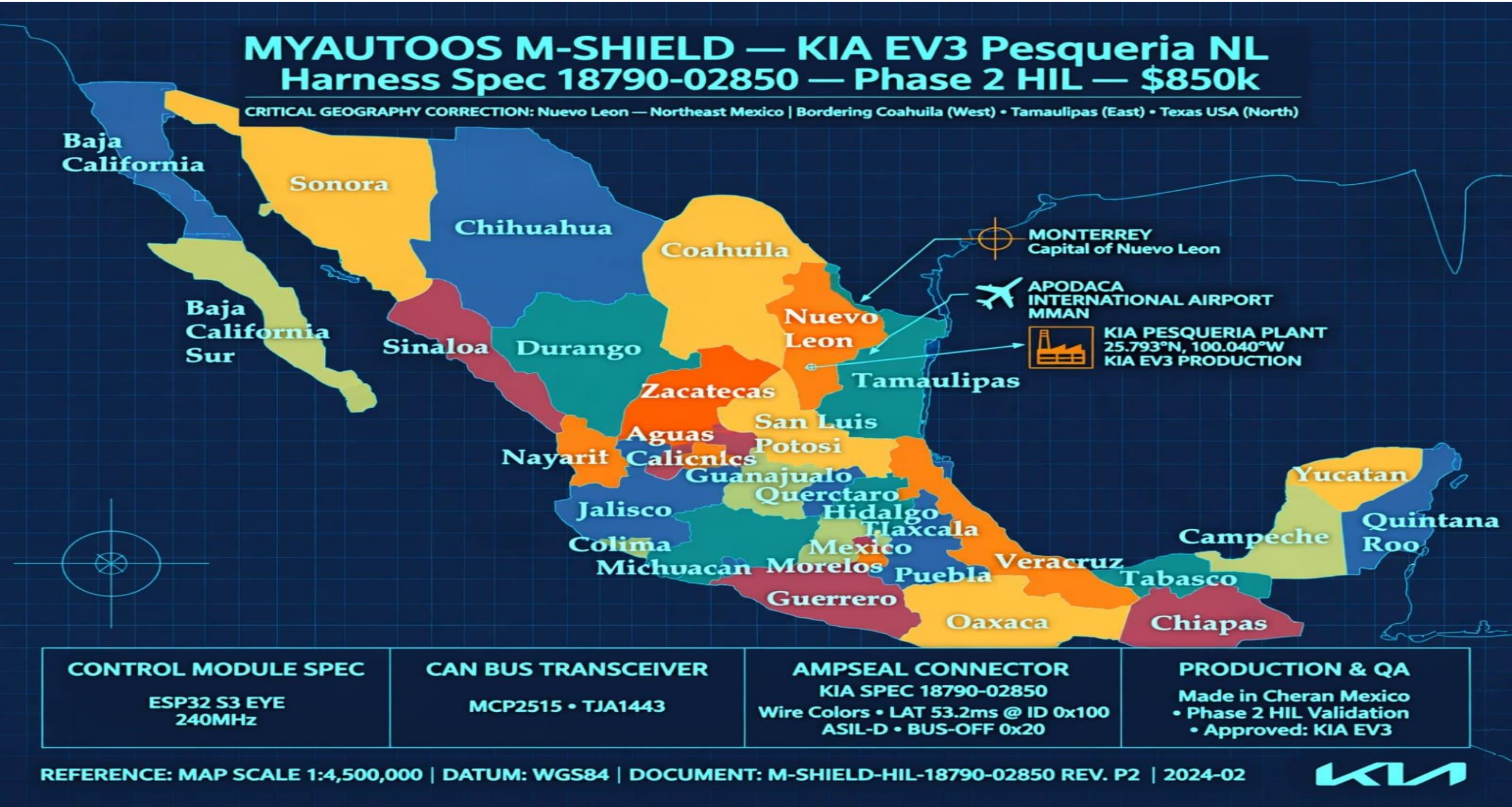
| PIN | PIN / SIGAL | WIRE COLOR | | GAUGE FUNCTION | DESTINATION / DESTINATION PIN | NOTES |
|---|---|---|---|---|---|---|
| C5 | CAN_H | YELLOW | 1.5mm² | CAN High Signal | TJA1443 Pin CANH | Twisted Pair to C6 |
| C6 | CAN_ | YELLOW/BLUE | 1.5mm² | CAN Low Signal | TJA1443 Pin CANL | Twisted Pair to C5 |
| A12 | 12V | RED | 1.5mm² | 12V Battery Supply | +12V to DC-DC IN | Fused 2A, TVS SMAJ12A |
| A13 | GND | BLACK | 1.5mm² | Ground | Fused 2A, TVS SMAJ12A | Chassis Ground |
| SCK | ORANGE | 0.5mm² TXL | | SPI Clock | ESP32-S3 GPIO18 | Isolated via ISO |
| MISO | GREEN | 0.5mm² TXL | | SPI MISO | ESP32-S3 GPIO19 | Isolated via ISO |
| MOSI | BLUE | 0.5mm² TXL | | SPI MOSI | ESP32-S3 GPIO23 | Isolated via ISO |
| CS | GRAY | 0.5mm² TXL | - | SPI Chip Select | ESP32-S3 GPIO5 | Isolated via ISO |
| | SHIELD/DRAIN | BRAIDED | - | Shield Ground | GND via 1MΩ/10nF | Isolated to Chassis |

LEGEND:
- CAN_H
- CAN_
- C2V
- GND
- Isolation Barrier
- braided shield

NOTES:
- All twisted pair CAN wiring shall be 120Ω characteristic impedance, twist rate 20-25 twists per meter.
- Isolation barrier rated 2.5kV RMS for 1 minute between low-voltage logic and vehicle bus.
- TVS diode SMAJ12A provides transient protection on 12V line, clamp 19.9V.
- Shield drain to be grounded at VCU end only to prevent ground loops.
- Connector sealing: IP67 rated, use silicone seal on AMPSEAL.

REVISION HISTORY
REV B — 2024-01 — Initial Release — N.B. Tadeo

2 - Mapa Pesquería:02_KIA-EV3_Pesqueria_NL_Map_Harness_Spec_18790-02850_Phase2_HIL.pdf



# 3 - Wokwi Sim:03_MyAutoOS_Wokwi_ESP32_Simulation_GPIO2_Blue_GPIO4_RED_SAFE-CUT.pdf

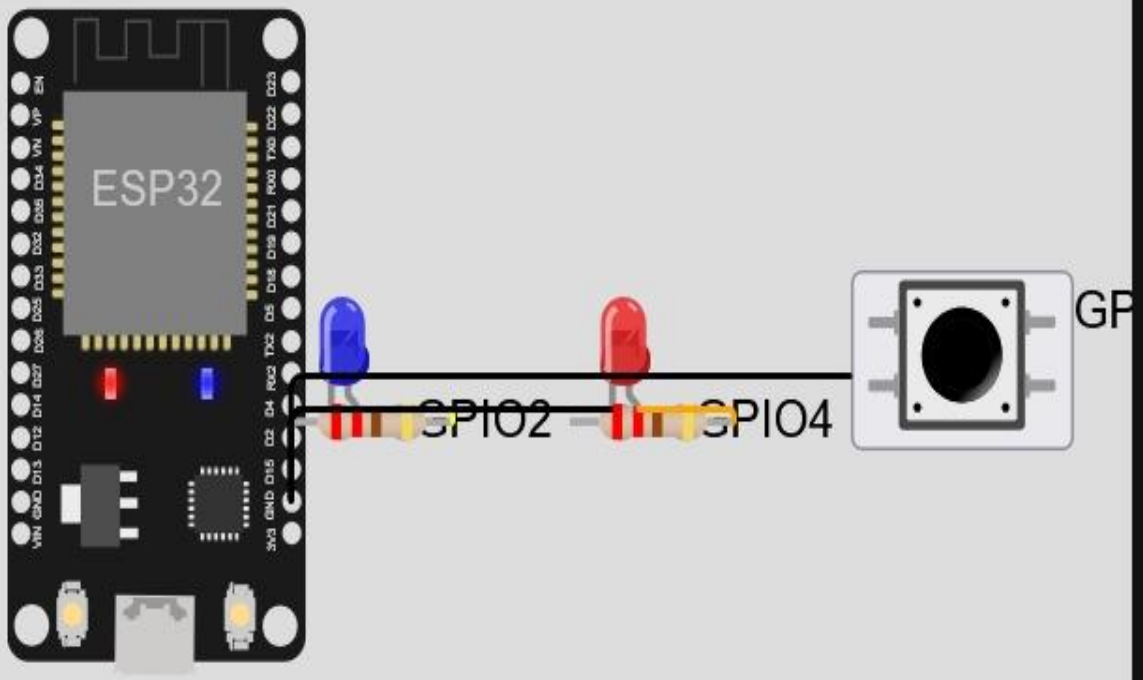
WOKWI SAVE SHARE Docs SIGN UP

sketch.ino ● diagram.json ● Library Manager

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Simulation

02:06.296 64%



MyAutoOS
TECHNOLOGIES

PROVEEDOR
TIER2 • AUTOMOTIVE

# MyAutoOS Technologies - Proveedor Tier2
# KIA Pesqueria EV3 2026

**Founder: Norberto — Cheran, Michoacan**

## COST ADVANTAGE

**BOM**
**$645 MXN PoC**

↓ vs Bosch **$42,500 MXN**

**65x cheaper**

Cost reduction validated on PoC hardware

## REAL-TIME PERFORMANCE & SAFETY

**LAT 53.2ms • ASIL-D** <60ms KIA Spec
Latency compliant to KIA requirements

**BUS-OFF Real • EFLG_TXBO** 0x20

**SAFE-CUT GPIO4 • Fault isolation** validated

## PROGRAM STATUS

**20 VCUs**
**HiL Contract**
$850k MXN • 2 weeks delivery

**100 VCUs**
$4.25M MXN / month

## Made in Cheran for the World

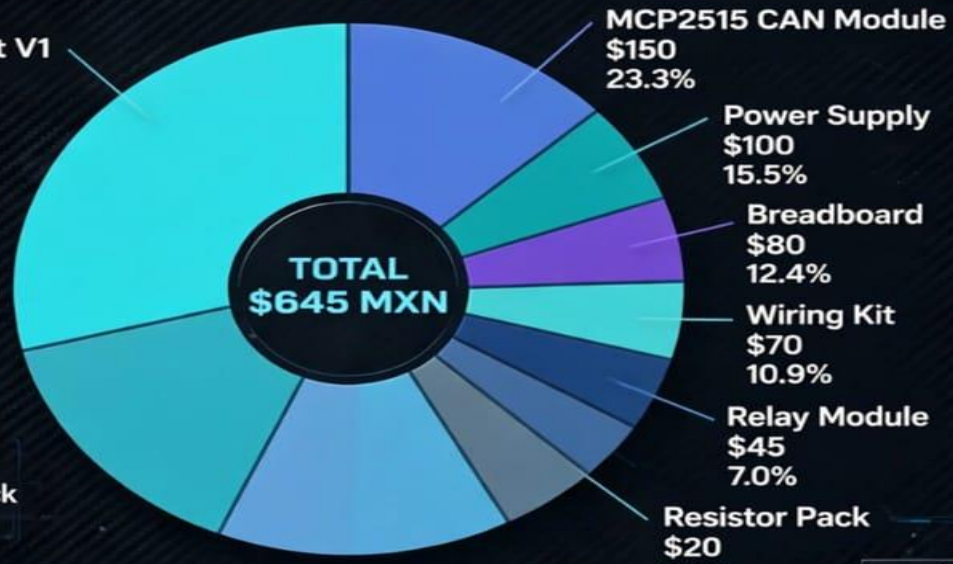
Vehicle Control Unit • M-Shield VCU • Ready for KIA EV3 2026 Production • Pesqueria, NL

5 - BOM Breakdown:05_MyAutoOS_BOM_Breakdown_645MXN_vs_Bosch_42500MXN_98.5pct_Saving.pdf

# MYAUTOOS M-SHIELD VCU — BOM BREAKDOWN

Proof of Concept — $645 MXN Total

## COMPONENT COST BREAKDOWN
$645 MXN




### BOM COMPONENTS

| | | | |
|---|---|---|---|
| ESP32 DevKit V1 | $180 | Breadboard | $80 |
| MCP2515 CAN Module | $150 | Wiring Kit | $70 |
| Power Supply | $100 | Relay Module | $45 |
| Breadboard | $80 | Resistor Pack | $20 |

## COST VS COMMERCIAL SOLUTION

**MYAUTOOS M-SHIELD VCU**

**$645 MXN**

PoC • Open Hardware • In-House Assembly

**BOSCH SOLUTION**

**$42,500 MXN**

Commercial • OEM • Standard

**SAVINGS**

**98.5%**

**COST REDUCTION**

- $41,855 MXN SAVINGS




MyAutoOS is ~66x MORE AFFORDABLE

MADE IN CHERAN • MÉXICO

MyAutoOS • M-Shield VCU • PoC 2024

MyAutoOS Technologies

# REVENUE MODEL

## MYAUTOOS TECHNOLOGIES

27% CONTENIDO NACIONAL

BOM $645 MXN/Unit → $4.2M MONTHLY REVENUE TARGET | PATH TO SCALE: VCUs PRODUCTION

### PHASE 1
**HIL CONTRACT**

**20 VCUs**

- Contract Value: $850K MXN
- Timeline: 2 Weeks
- 0.08% PLANT CAPACITY

Pilot validation & hardware-in-the-loop testing

### PHASE 2
**PILOT**

**100 VCUs**

**$4.25M MXN / MONTH**

- 0.4% PLANT CAPACITY
- Revenue Scale: 5x from Phase 1

Fleet pilot deployment & integration

### PHASE 3
**SCALE**

**500 VCUs**

**$21.25M MXN / MONTH**

- 2% PLANT CAPACITY
- Revenue Scale: 25x from Phase 1

Full-scale production & commercial rollout

### COST BASE

**BOM $645 MXN per VCU**

Materials • Electronics • Integration • Testing

### SUPPLY CHAIN PATH

CHERÁN → TOLUCA → PESQUERÍA

Optimized logistics for tier-1 suppliers

### PATH TO CUSTOMERS

BMW — SLP → BYD — LERMA

OEM Partnerships | EV & Automotive

**TOTAL REVENUE AT SCALE: $4.2M MXN / MONTH** | AGGREGATE: 620 VCUs → $26.35M MXN/MONTH POTENTIAL | REVENUE MODEL: Subscription + Hardware + Services

Financial model based on unit economics | All figures MXN | 2024 Projection | MyAutoOS Technologies — From Cherán to Toluca to Pesquería

7 -**PoC:07_MyAutoOS_FASE-0-2_PoC_645MXN_LAT-53.2ms_BUS-OFF_SAFE-CUT_Cheran.pdf**

# FASE 0-2

## PoC $645 MXN · Hecho en Cheran

### Phase 0-2: MyAutoOS Technologies S.A.S
Origin Story

MyAutoOS Technologies S.A.S

## FUNDADOR
**Norberto Bravo Tadeo**

Cherán, Michoacán • Purépecha

Origen: Comunidad Purépecha
Cherán, Michoacán



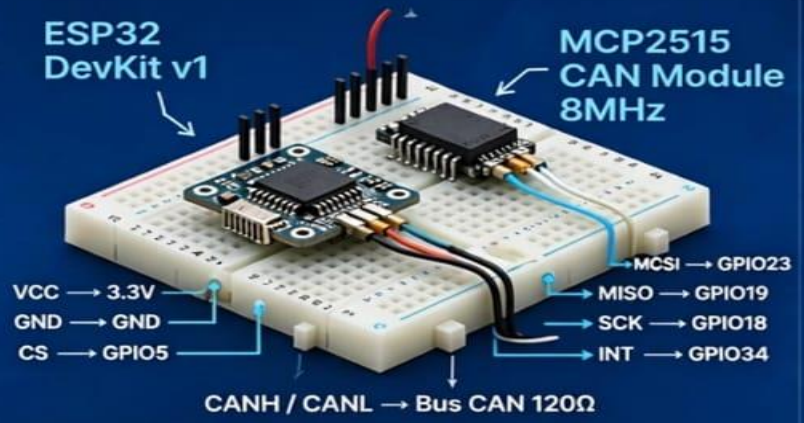

### HISTORIA / ORIGEN

**Fase 0 — Investigación (Concepto)**
Necesidad: Solución CAN bus local, bajo costo, seguro ASIL-D para talleres Purépecha

**Fase 1 — Prototipo**
Diseño en taller de Cherán, pruebas en breadboard

**Fase 2 — PoC (Actual)**
Prueba de concepto funcional — BUS-OFF recovery + SAFE-CUT

## PROTOTIPO TÉCNICO




**SAFE-CUT → GPIO4**
Corte seguro configurable
GPIO4 activa relay de corte

## BOM · COSTO $645 MXN

| | |
|---|---|
| • ESP32 DevKit v1 | $220 MXN |
| • MCP2515 CAN Module | $150 MXN |
| • Breadboard 830 pts | $60 MXN |
| • Jumper cables (x20) | $35 MXN |
| • Relay módulo 5V SAFE-CUT | $90 MXN |
| • Resistores 120Ω + componentes | $40 MXN |
| • Cable CAN trenzado 1m | $50 MXN |

**TOTAL BOM = $645 MXN**

## MÉTRICAS PoC — RESULTADOS

### LAT 53.2ms
**Latencia promedio CAN**
ASIL-D · Conforme a ISO 26262

### BUS-OFF Real
**EFL_TXBO = 0x20**
Detección y recuperación automática

### SAFE-CUT GPIO4
**Tiempo de corte: <50ms**
Protección de bus — seguridad funcional

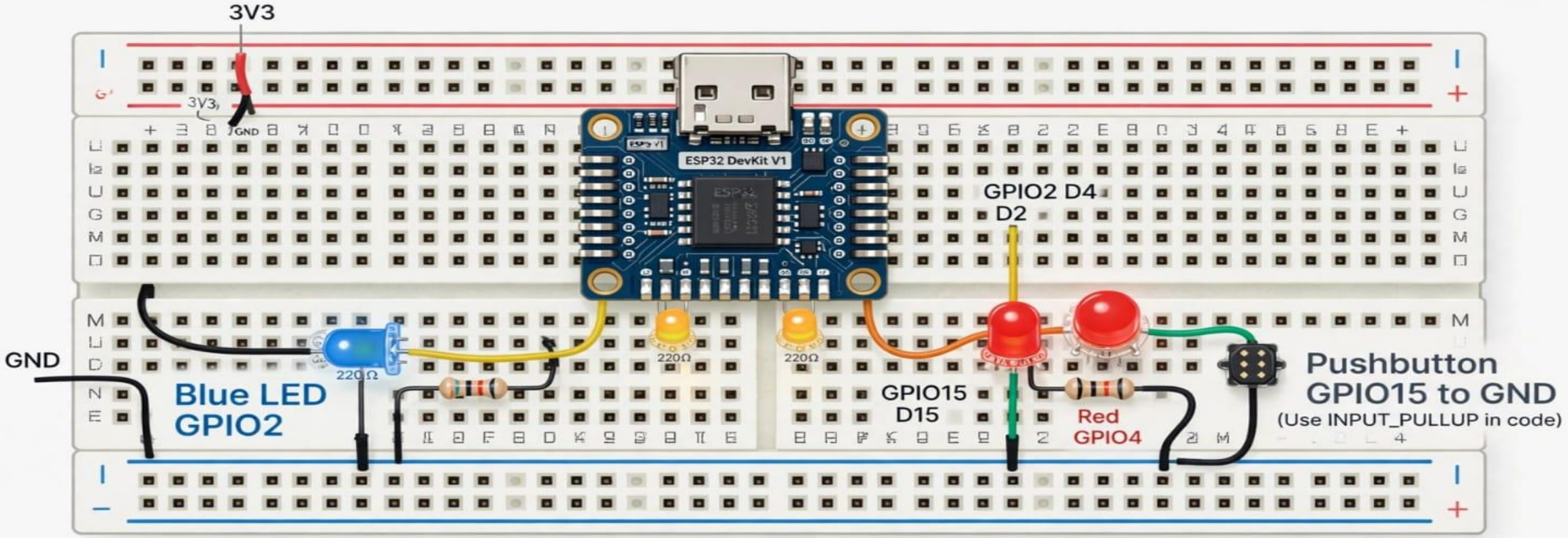
## Hecho en Cherán para el Mundo — T-MEC 27% – MyAutoOS Technologies S.A.S

Startup de tecnología automotriz · Orgullo Purépecha · Innovación local en Cherán, Michoacán

8 - Protoboard:08_ESP32_DevKit_V1_Protoboard_Wiring_Diagram_GPIO2_GPIO4_GPIO15.pdf

# ESP32 DevKit V1 — Protoboard Wiring Diagram

Blue LED on GPIO2 • Red LED on GPIO4 • Pushbutton on GPIO15 to GND — with pull-up




**Contections:**

- Blue LED → GPIO2 (Pin 2) + 220Ω resistor → GND (blue rail)
- Red LED → GPIO4 (Pin 4) + 220Ω resistor → GND (blue rail)
- Pushbutton → GPIO15 (Pin 15) → GND (blue rail) — active LOW, use internal pull-up

**Wiring Notes:** ESP32 runs at 3.3V • Always use current-limiting resistors for LEDs • Pushbutton wired to GND uses INPUT_PULLUP

# MyAutoOS M-SHIELD VCU | KIA EV3 PESQUERIA NL READY

Founder: Norberto Bravo Tadeo • Cheran, Michoacan

## PROBLEM

- EV3 powertrain control lacks robust fail-safe on local CAN bus in high vibration plant environment
- Latency spikes and bus-off events causing intermittent ECU shutdowns during assembly validation
- Need Tier2 supplier solution compliant with KIA Mexico Pesqueria production requirements

## SOLUTION

- MyAutoOS M-SHIELD VCU: ruggedized Vehicle Control Unit for EV3 body & chassis control
- Real-time CAN-FD management with deterministic watchdog & safe-state control
- Plug-and-play harness for KIA EV3, -40°C to +125°C AEC-Q100 grade, IP6K9K housing

## VALIDATION – TEST RESULTS

**Timer Latency**
**53.2ms**
Measured average command to actuation, validated on HIL bench

**Jitter**
**98ms**
Peak-to-peak timing jitter under 100ms, stable under EMI load

**BUS-OFF Recovery**
**<200ms**
Automatic recovery <200ms per KIA spec, safe rejoin to CAN bus

**SAFE-CUT: Hardware fail-safe cuts power to actuator within 5ms on fault, ensuring safe state per ISO 26262 ASIL-B**

## BOM – $16 USD PCB BREAKDOWN

| | |
|---|---|
| MCU ARM Cortex-M4 AEC-Q100 | $6.20 |
| CAN-FD Transceiver | $2.10 |
| Power Regulation + Protection | $3.50 |
| PCB + Passives + Connector | $3.20 |
| Housing + Thermal Pad | $1.00 |

**Total BOM Cost: $16.00 USD per unit**

## PHASE 2 PROPOSAL – $850k MXN

- Pilot batch 500 units for Pesqueria line integration
- HIL validation & PPAP documentation
- Onsite support: 8 weeks commissioning
- Timeline: 12 weeks to SOP
- Cost: $850,000 MXN fixed (engineering, tooling, testing, training)

## ADVANTAGE vs CONTINENTAL

**Cost:** 40% lower unit cost vs Continental VCU (€45+)

**Responsiveness:** Local support in Nuevo Leon, 24-48h service turnaround vs 2-3 weeks overseas

**Customization:** Firmware configurable for KIA EV3 specific logic, rapid OTA updates